Journal of Development Economics 165 (2023) 103173

Contents lists available at ScienceDirect

Journal of Development Economics

journal homepage: www.elsevier.com/locate/devec

Regular article

Mitigating agricultural fires with carrot or stick? Evidence from China ☆

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ARTICLE INFO

Dataset link: https://doi.org/10.1016/j.jdeveco.2023.103173

JEL classification:
H23
Q12
Q52
Q53

Keywords:
Agricultural fire
Environmental externality
Straw burning
Biomass power plant

ABSTRACT

This paper examines the effects of biomass power plants (BPPs) on farmers' use of agricultural fires for land clearance in China from 2001 to 2019. We show that the entry of BPPs leads to a significant reduction of agricultural fires by 14%. Farmers near BPPs display stronger responses, leading to a more significant reduction in straw burning, especially during high agricultural fire seasons. The notable decline in agricultural fires is likely driven by economic incentives provided by BPPs to farmers for collecting crop straw from their land. Additionally, straw-burning bans have limited effectiveness in reducing total agricultural fires, but they appear to reduce straw burning during nighttime, when the monitoring of agricultural fires is easier. We also provide evidence that local air quality has markedly improved, resulting in substantial health benefits that surpass the social benefits of reducing carbon emissions.

1. Introduction

Technological advances have greatly improved monitoring of environmentally destructive behavior by large polluters. Many countries, including the US, China, and India, have built sophisticated high-frequency monitoring systems that focus on large firms and keep their emissions of air and water pollutants under close government surveillance. These systems directly reduce the manipulation of reported pollution by polluters and government officials (Karplus et al., 2018; Greenstone et al., 2022), and also help discipline polluters' environmentally destructive behavior and improve their environmental performance (Zhang et al., 2018; Zou, 2021). However, environmental externalities from smaller contributors, especially from households, are much harder to monitor. Closely monitoring their behavior is highly costly due to the large number of potential contributors and their damaging practices usually continue regardless of government prohibitions (Aunan et al., 2019; Rao et al., 2021). How to effectively regulate their environmental externalities remains a challenge to governments.

This paper focuses on a specific environmental externality caused by rural residents, i.e., open agricultural burning. In many countries,

☆ We are thankful to the Research Center for Green Economy and Sustainable Development and Institute for Global Development of Tsinghua University, Environmental Defense Fund China, and Harvard-China Project on Energy, Economy and Environment. This paper is supported by the National Key Research and Development Program of China, China (2020YFA0608603), the National Natural Science Foundation of China (72134006, 72243007, 72304272), and the Chinese Universities Scientific Fund, China (15053362).

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[1] For example, in Punjab and Haryana of India, stubble burning accounts for 30% to 40% of air pollution in October and November. Source: https://www.reuters.com/world/india/india-struggles-put-out-crop-waste-fires-that-fuel-air-pollution-2021-11-11/.

[2] China has partially banned agricultural open burning since 1997. In the US, Oregon prohibited this practice in 2009, and many other states impose restrictions on farmers by requiring them to obtain permits for open burning.

[3] The US National Aeronautics and Space Administration (NASA) provides real-time monitoring on open burning, within one minute of satellite observation, but data are currently available only for the US and Canada. Source: https://wiki.earthdata.nasa.gov/pages/viewpage.action?pageId=258343755.

[4] Source: http://sthjt.gxzf.gov.cn/zwxx/qnyw/t5493486.shtml, http://env.people.com.cn/n1/2018/0822/c1010-30244953.html.

[5] For example, Shijiazhuang, a city in Hebei province of China, installed a monitoring system with 529 cameras for agricultural fires in farmland near railways and highways, with an annual cost around 30 million Chinese Yuan (CNY hereafter, 1 CNY≈ 0.14 USD). Source: http://www.sjz.gov.cn/download.jsp?pathfile=/yjsgk/atm/1613639344654/20210222150033124.pdf.

https://doi.org/10.1016/j.jdeveco.2023.103173
Received 26 March 2022; Received in revised form 14 July 2023; Accepted 14 August 2023
Available online 22 August 2023
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including China, India, and Brazil, farmers use open fires to dispose of crop residues (straw stubble) as a cheap way to clear their land. The combustion of straw stubble produces substantial amounts of air pollutants, such as particulate matter and sulfur dioxides, leading to a rapid and severe deterioration in local air quality.[1] These air pollutants pose large risks to human physical and mental health by increasing mortality of infants and the elderly, as well as impairing cognitive abilities (Rangel and Vogl, 2019; He et al., 2020; Graff Zivin et al., 2020). This practice is therefore restricted or even strictly prohibited by many governments.[2] However, governments face high costs in closely monitoring agricultural fires. Remote sensors on satellites can capture hotspots caused by open burning from space, but real-time satellite monitoring is rarely available in most countries.[3] The delay in monitoring causes obstacles for government officials to arrive at fire sites promptly and find out (or prove) who sets the fires. Therefore, satellite monitoring is mostly used for ex-post evaluations of the severity of open burning, but cannot directly help governments strengthen enforcement. To better monitor agricultural fires, some local governments in China have recently installed real-time cameras with infrared sensors that can automatically detect open burning and store visual records for evidence.[4] However, such monitoring systems are highly costly and hence are not widely used.[5] Due to these obstacles in monitoring, the bans on agricultural open burning in many countries are impeded by weak enforcement.

This paper studies an alternative approach to solving this problem, namely through biomass power plants (BPPs). BPPs remunerate farmers for collecting crop straw from their land and use the acquired straw to generate electricity. The payments from BPPs for crop straw internalize the environmental costs of farmers' straw-burning practice à la Coase. The effectiveness of such payments in mitigating agricultural fires depends on whether they are sufficiently large to cover farmers' private costs of straw collection, which may vary substantially across different individuals. BPPs collect straw not only from farmers who would burn straw if not for the remuneration, but also from farmers who would dispose of straw in environmentally friendly ways, such as using straw as food for livestock or fertilizer for crops. These farmers may have lower private costs of collecting straw—otherwise they would have burned the straw to clear their land. These lower costs imply that the acquiring price of straw could still be lower than the straw-collecting costs of farmers who choose to use open burning to dispose of straw. Therefore, the payments from BPPs could fail to internalize the external environmental costs of open agricultural burning and so do not necessarily mitigate agricultural fires.

We use the entry of BPPs in China as a quasi-natural experiment to examine their impact on the occurrence of agricultural fires detected by satellites from 2001 to 2019. To address the challenge of endogenous site selection, we compare BPPs in operation to untreated BPPs, including those in the process of preparation and a small number in the withdrawal process. Typically, it takes an average of 3.5 years from business registration for BPPs in China to obtain government approvals, complete construction of plants and transmission lines, commence operations with electricity supply licenses. The lengthy establishment process, coupled with variations in the timing of business registrations, creates plant-specific differences in entry time. We explore these variations to uncover the causal impact of BPP entry on reducing agricultural fires.

We find that on average, the entry of a BPP statistically significantly reduces nearby agricultural fires by 14%, with larger reductions in regions closer to BPPs (as much as −23%) and in high agricultural fire seasons (−17%), i.e., the periods before sowing and after harvest. Our evidence suggests that economic incentives, provided by BPPs to acquire crop straw from farmers, are a likely channel for the significant reductions in agricultural fires. We find that the reductions are mostly driven by BPPs that are not financially constrained in remunerating farmers for straw collection and BPPs that are able to provide farmers higher payments due to receipt of production subsidies from government. BPPs with greater demand for crop straw, reflected by their electricity output and capacity utilization, also exhibit larger effects on reducing agricultural fires.

Additionally, we compare the impact of BPPs to that of straw-burning bans. We find that straw-burning bans in China have a much smaller effect on reducing agricultural fires compared to BPPs. Although the bans appear to significantly reduce agricultural fires at night when fires are easily visible from a long distance, the number of nighttime fires is only one-ninth of the number of daytime fires, resulting in limited overall reduction in total fires.

Finally, we show that BPP entry has a negligible effect on reducing vegetation cover but significantly improves air quality. Back-of-the-envelope calculations indicate that public health benefits due to reductions in agricultural fires well exceed the monetized social benefits of carbon emissions reductions from BPPs' generated electricity and even slightly exceed the private revenues of BPPs from electricity sales.

This paper contributes to several strands of literature. First, this work is linked to existing literature on the escalation and mitigation of agricultural fires. Straw burning is a labor-saving practice used by farmers to clear their land so that their sowing and harvesting activities can be less labor-intensive (Garg et al., 2022). Hence, the opportunity cost of labor has a determining effect on the use of agricultural fires. For example, Behrer (2019) finds that an anti-poverty program in India significantly increased cropland fires because the program increases agricultural wages and induces farmers to dispose of crop residues with the labor-saving straw burning. Garg et al. (2022) show that access to rural roads also increases agricultural fires because it leads to reductions in the farm labor workforce. This paper contributes to the literature by showing that the entry of BPPs, which remunerate farmers for their labor costs of collecting straw, has a significant effect on reducing agricultural fires.

This paper is closely related with Nian (2023), which also investigates the impact of BPP entry on agricultural fires in China. However, Nian (2023) adopts an "inner ring versus outer ring" strategy by comparing agricultural fires within 10 km of BPPs (including fires within 5 km of BPPs and fires between 5 and 10 km from BPPs) to fires between 10 and 15 km away from BPPs. This strategy not only requires additional assumptions beyond quasi-random variations in the timing of treatment (Pollmann, 2023), but the chosen 10 km cutoff is substantially smaller than the industry's consensus on the operational radius of BPPs, potentially causing the outcomes in the outer ring to be influenced by the treatment.[6] In contrast, our paper compares agricultural fires near operational BPPs to fires near counterfactual locations where the treatment could have occurred. This type of approach typically relies on weaker identification assumptions and demonstrates high robustness against omitted variable biases (Borusyak and Hull, 2020; Pollmann, 2023).

In addition, our comprehensive data on BPPs, including information on their operations and financial status, allows us to closely investigate the underlying mechanisms behind the reduction in agricultural fires. We provide suggestive evidence that the observed reduction in agricultural fires is likely driven by economic incentives provided by BPPs to farmers. Previous studies have utilized randomized controlled trials and government programs to examine the effectiveness of economic incentives, such as payments for ecosystem services, in forest protection (Alix-Garcia et al., 2015; Jayachandran et al., 2017; Ferraro and Simorangkir, 2020) and reducing agricultural fires (Edwards et al.,

[6] For instance, according to a newspaper report dated November 24, 2020 by the China Environment News, which is under the administration of the Ministry of Ecology and Environment, BPPs in China typically operate within a radius of no more than 100 km, which is ten times greater than the 10 km cutoff. Source: http://epaper.cenews.com.cn/html/2020-11/24/content_99720.htm.

2020; Jack et al., 2022). In particular, Edwards et al. (2020) find that the economic incentives in their randomized field experiment seem too small to induce farmers to reduce agricultural fires. Jack et al. (2022) show that economic incentives with upfront payments significantly reduce crop residue burning whereas standard incentives with no upfront payments have null effect, possibly due to distrust of standard incentives. Consistently, we find larger reductions in agricultural fires around the BPPs that are able to provide farmers with greater economic incentives, namely the BPPs with production subsidies from the government and BPPs with higher electricity capacity utilization and output. Additionally, we observe a negligible effect for BPPs that do not fulfill their legal obligations regarding debt payoff, likely due to smaller economic incentives provided by these plants or farmers' lack of trust in their promise of payment.

This paper is also relevant to the literature on control of open burning. Fires are widely used for land clearing, and this practice generates not only air pollutants that damage human health but also substantial carbon emissions.[7] Balboni et al. (2021) discuss different policies to control forest fires set by firms in Indonesia and show that property rights reforms have a negligible effect on reducing fires, whereas increasing punishments for setting fires can reduce fires by up to 80%. Nevertheless, forest fires are usually much larger in scale than agricultural fires and are easier for governments to detect, whereas most agricultural fires in China are set by rural households rather than firms, suggesting difficulty for governments in monitoring their illegal behavior.

Furthermore, incomplete environmental regulation may improve compliance in emissions of regulated pollutants, but provoke types of polluting behavior that are less regulated, such as crop residue burning (Hernández-Cortés, 2022). Ashraf et al. (2016) compare the effectiveness of subsidies and punishments in addressing externalities with a theoretical model and two cases from New York and Zambia. They show that when judicial strength is low, subsidies are more effective than fines because effective sanctions are impossible if inspectors cannot determine who is responsible for the externalities. The present paper echoes their findings by showing that BPPs significantly reduce straw burning activities that are hard to monitor, but punishment imposed by straw-burning bans has a limited effect except on straw burning at night, when inspectors can more easily identify fires.

Last, this paper is relevant to the literature on co-benefits of reducing carbon emissions (Fullerton and Karney, 2018; Scovronick et al., 2019; Aldy et al., 2021). BPPs are thought to contribute to achieving net-zero carbon emissions, but are also controversial due to fears of deforestation (see Favero et al., 2020). However, China maintains strict regulation on forests and BPPs in China use agricultural residues as fuel. Accordingly, we find no effect of BPP entry on vegetation cover, but sizeable health benefits due to the reduction in agricultural fires.

The remainder of this paper proceeds as follows. Section 2 introduces agricultural fires and regulations on straw burning in China and also provides background on China's biomass power plants. Section 3 describes the empirical strategy. Section 4 presents the main results. Section 5 compares the effects of BPPs with those of straw burning bans. Section 6 discusses the benefits and costs of BPPs. Section 7 concludes.

2. Background

2.1. Agricultural fires and regulations on straw burning

Wildfires in China mainly occur on farmland rather than woodland or grassland. In recent decades, rural households in China have switched from cooking and heating with wood and straw to more efficient alternatives such as electricity, natural gas, and coal (Tao et al., 2018). This energy transition by rural households is associated with increased agricultural fires as surplus agricultural waste is burned. Fires in farmland, detected by the Moderate Resolution Imaging Spectroradiometer (MODIS) fire product, increased by more than 120% from 2003 to 2019, and fires in woodland decreased by about 60% in the same period (see Fig. 1).[8] In 2019, the number of agricultural fires recorded by the MODIS fire product was almost 2.5 times greater than the number of forest fires.

China has officially restricted straw burning for more than 20 years. The first straw burning ban was introduced in 1997, when the Ministry of Agriculture issued an administrative order prohibiting straw burning in the summer harvesting season. The administrative order was written into law in 2000 when the National People's Congress approved the amendment of the Atmospheric Pollution Prevention and Control Law. The law stipulates that straw burning is prohibited in areas that are densely populated or close to airports or arterial roads. At that time, mitigation of agricultural fires mainly relied on mandatory regulations such as these. Temporary measures were implemented around 2008 to further enhance air quality in Beijing during the Olympic Games (Chen et al., 2013; He et al., 2016).

However, despite these government regulations, agricultural fires continued to increase rapidly, particularly after 2009 (see Fig. 1). The surge in agricultural fires was halted when the central government promoted the utilization of crop straw in 2013. Local governments were ordered to use "carrot and stick" methods to reduce agricultural fires. They treated companies that recycle straw as the main contributors to reducing agricultural fires and supported them with preferential policies. They also enhanced the implementation of straw burning bans, and some provinces even passed legislation completely prohibiting straw burning across their entire jurisdiction. Nowadays, farmers who burn crop straw illegally will be fined 500 to 2000 CNY (about 70 to 300 USD) and may even be detained for five days or more, according to several Chinese laws.

2.2. Biomass power plants in China

Biomass power plants typically consume crop straw, animal waste, residential garbage, and wood to generate electricity. We focus on agricultural and forestry BPPs that use crop straw and wood as fuel. For convenience, we refer to these simply as BPPs throughout the paper.

The first BPP in our sample came into operation in 2006, and the biomass power sector began growing rapidly in 2009. In the decade from 2009 to 2019, the sector's installed capacity grew tenfold from 1088 megawatts to 10,920 megawatts. The amount of electricity generated follows a similar trend, implying almost constant operating hours for biomass power plants' generators (4405 h on average in 2019). These operating hours are close to those of coal-fired power plants (4376 h in 2019) and are substantially higher than those for other renewable energy sources, such as hydropower (3637 h), wind power (1938 h), and solar power (1097 h).

Despite the biomass power sector's rapid growth in the last decade, establishing a BPP is a lengthy process. Fig. D1 in the Online Appendix illustrates the main procedures involved. The process begins with the company registering the business with the Administration

[7] Hsiao (2021) shows that CO_2 emissions due to forest fires set by the palm oil industry in Indonesia and Malaysia accounted for 4.7% of global emissions from 1986 to 2016.

[8] The substantial reduction in woodland fires detected by MODIS is consistent with the official statistics of China. According to China Statistical Yearbooks, the annual number of forest fires in China has reduced from 10,463 in 2003 to 2345 in 2019. The Chinese government attributes this reduction to the substantial increase in resources allocated to forest fire prevention. From 2008 to 2015, the number of forest firefighters increased by 53.8% to 113,000 persons. Government expenditure on forest fire prevention amounted to 30 billion CNY (about 4.3 billion USD) during that period. (Source: National Forest Fire Prevention Plan (2016–2025), http://www.gov.cn/xinwen/2016-12/29/content_5154054.htm.)

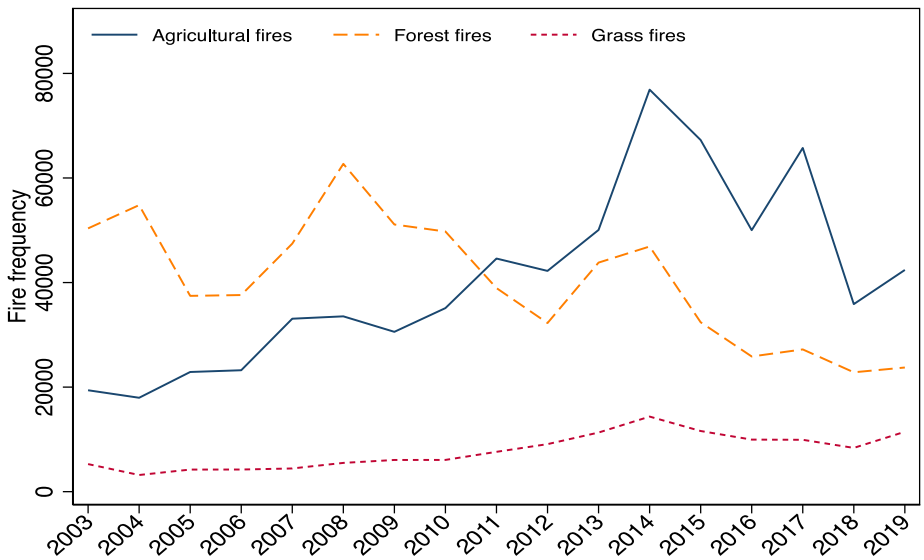


Fig. 1. Frequency of agricultural fires, forest fires, and grass fires in China.
Notes: Agricultural fires, forest fires, and grass fires are identified as hotspots detected by MODIS in farmland, forest, and grassland, respectively.

for Market Regulation, followed by obtaining approvals from at least six provincial-level government departments.[9] Each department typically has a maximum of 20 working days to review and make a decision on the company's application. Some departments also require the application to be published on their websites for an additional 20 days before granting approval. Furthermore, the company cannot directly submit applications to the provincial-level departments but must instead submit them to local authorities, who then forward them to the provincial governments after conducting initial reviews. As a result, this bureaucratic process often takes a minimum of 6 months. In cases where the application is denied by any department, the company must revise and resubmit it for reconsideration, further prolonging the process. Additionally, after obtaining approvals from the government departments, the company typically needs to purchase land use rights through land auctions organized by the Bureau of Planning and Natural Resources. This step can be a time-consuming process that may extend over several months or even years, particularly when it involves land acquisition from residents and the demolition of existing housing structures.

After completing all these procedures, the company goes to the Provincial Development and Reform Commission to apply for planning and construction permits for the project. During construction, the company must apply to the provincial power grid company for connection to the electricity grid, which then requests grid-planning institutes to design a power connection plan and hires construction forces to build transmission lines. After finishing plant and transmission line construction, the company must perform a trial run before starting project acceptance procedures, and after completing these procedures the company can finally apply to the Provincial Energy Administration for an electricity business license, which is a requisite for power plants to supply electricity to power grids.

The approval process for projects in the energy sector can be complex and time-consuming, with outcomes sometimes affected by various factors. Historically, the energy sector in China faced challenges with transparency, particularly in the 2010s when some government officials were implicated in corruption scandals related to project approvals. For instance, in 2014, several key figures in energy-related government departments were arrested on charges of bribery and corruption, including Tienan Liu, the former director of the National Energy Administration.[10] Consequently, even if a biomass power company identifies a suitable location for electricity generation using crop straw, it may encounter difficulties in obtaining government approval. Therefore, the successful acquisition of government approval and the duration of the approval process may sometimes depend not only on adhering to required protocols but also on the company's connection with relevant authorities.

In some cases, high uncertainty could also arise from unforeseen events involving BPPs and third parties. For instance, in Shaanxi province, one BPP successfully obtained approval from the PDRC in 2008 but failed to complete factory construction by the end of 2022 due to a legal dispute with a construction company.[11] Similarly, in Sichuan province, a BPP faced financial difficulties during plant construction and was unable to fully remunerate the construction company. Although the plant construction was completed, the BPP never went into operation.[12] Additionally, delays in the process can also be caused by grid companies. For instance, in another BPP case in Sichuan, the plant was ready for operation, but the grid company did not build transmission lines because the construction cost of transmission lines was considered to exceed the future revenues for the grid company.[13]

Consequently, the unpredictability associated with establishing biomass power plants is primarily determined by the relationship between BPPs and the government, as well as unforeseen incidents involving third parties. These factors are generally unrelated to the straw-burning activities of local farmers. Furthermore, in Section 3.3, we demonstrate that registered BPPs, both with and without electricity business licenses, exhibit similar characteristics in terms of local natural and economic conditions. These characteristics include factors such as nearby agricultural fire frequency, biomass resource availability,

[9] Once completing business registration, the biomass power company submits a project proposal to the Provincial Development and Reform Commission (PDRC). The PDRC then provides the company with detailed instructions. The company must obtain approvals from the Department of Housing and Urban–Rural Development, the Department of Natural Resources, the Department of Water Resources, and the Department of Ecology and Environment.

[10] Source: http://fanfu.people.com.cn/n/2014/1210/c64371-26184398.html.

[11] Source: http://sndrc.shaanxi.gov.cn/fgwj/2008nwj/1016307z2aQvy.htm, https://aiqicha.baidu.com/wenshu?wenshuId=b4bef5718228cc328dc9ff43648d62c29e8c038c.

[12] Source: https://aiqicha.baidu.com/nwenshu?wenshuId=5b96bdd3a83e55ba3b1b8cf523db7dc8337eb869.

[13] Source: http://finance.sina.com.cn/g/20090728/15096538746.shtml.

distances to major infrastructures, and county-level economic variables. Therefore, we draw upon the extensive procedures and uncertainties involved in the process from business registration to obtaining an electricity business license to identify the causal effect of BPP entry on agricultural fires.

3. Empirical design

3.1. Empirical specification

We use a staggered event-study design to exploit variations in timing of BPP entry. However, comparison of regions with and without BPPs would be susceptible to endogeneity problems due to the non-random site selection of power plants. For example, counties with severe agricultural fires generally have greater biomass resources and hence larger incentives to establish BPPs.

To overcome this problem, we exploit the lengthy process of establishing BPPs, as described in Section 2.2, and use as the untreated group plants that are in the process of preparation or withdrawal, i.e., plants that have registered their business but have not entered the market by the end of the sample period. The long duration and high unpredictability in establishing BPPs not only increases variation in the timing of BPP entry but also creates untreated units.[14] We use these variations to identify the effects of BPP entry on agricultural fires.

More specifically, we consider the following event-study specification:

$$Fire_{iym} = \beta Biomass_{iym} + \alpha X_{iym} + \gamma_{im} + \eta_{pym} + \epsilon_{iym} \tag{1}$$

where $Fire_{iym}$ is the frequency of agricultural fires surrounding plant i in year y and month m, $Biomass_{iym}$ is a dummy variable that equals one if plant i is in operation in year y and month m, and zero otherwise, X_{iym} is a set of weather control variables, including temperature, precipitation, wind speed, wind direction, and relative humidity, γ_{im} represents plant-month fixed effects, η_{pym} represents calendar-month-specific province-time fixed effects, and ϵ_{iym} is the error term.

Eq. (1) is a standard staggered event-study design. $Biomass_{iym}$ compares the treated plant i to untreated plants (plants that have not come into operation by the end of the sample period), treated plants that are not yet in operation, and other treated plants that have come into operation (Goodman-Bacon, 2021). This design is subject to the problems of negative weights (de Chaisemartin and d'Haultfoeuille, 2020; Goodman-Bacon, 2021). We discuss the identification issues in detail in the Online Appendix A.

Although plants in operation and in the process of preparation or withdrawal could share similar characteristics regarding agricultural fires in their regions, we must still consider potential confounding factors occurring in parallel with BPP entries. We address this problem by controlling for calendar-month-specific province-time fixed effects η_{pym}. Controlling for time-varying province fixed effects means that our model uses within-province variations to identify the effects of BPP entry. The fixed effects absorb all variations at the province level and higher levels, such as seasonal variations in agricultural activities across different provinces of China, the subsidy policy for straw utilization in ten provinces discussed by He et al. (2020), and the straw burning bans discussed in Section 6. In addition, we control for month-specific plant fixed effects γ_{im} to allow for idiosyncratic seasonal patterns of agricultural fires surrounding each BPP.

There could also be other confounding factors at a more granular level. For example, in some counties, the local government might strengthen the enforcement of straw burning bans and at the same time provide favorable policies for straw utilization, which could promote the creation of BPPs. However, due to the long time required to establish BPPs, such policies are unlikely to be concurrent with the timing of BPP entries, and are more likely to occur before BPP entries. Therefore, they can be tested by estimating dynamic treatment effects with lags and leads.

To count agricultural fires surrounding each BPP, we must choose a radius. He et al. (2020) count the fire frequency within 50 km of the center of each county and note that the average county radius is 33 km (calculated based on the average county area). The impact of one BPP could span several nearby counties if straw supply from the county where the BPP is located cannot satisfy its demand for electricity generation. Because the actual operational scope of the BPPs is unobservable to us, we start the empirical analysis with a 10 km bin and count agricultural fires in rings with distance between $r-10$ km and r km from each plant, where $r = 10, \ldots, 120$. The maximum distance of 120 km is slightly larger than the biomass power industry's consensus that BPPs collect straw within a distance of 100 km at most. The further a ring is from the BPP, the higher the transportation costs the BPP will face. Therefore, we expect that the impact of BPP entry decreases with distance from the plant.

We estimate Eq. (1) with ordinary least squares (OLS) and Conley spatial standard errors. To be conservative, we choose a large bandwidth for the Conley standard errors (600 km), which is five times the largest ring we consider. The robustness checks in the Online Appendix A consider alternative bandwidths (200, 500, and 1000 km) for the Conley standard errors and also clustered standard errors at the plant level. Because the dependent variable, agricultural fire frequency, is a count variable, we also consider a Poisson regression for estimation in the robustness checks.

3.2. Data

The fire data we use are from the MODIS units aboard two satellites (Terra and Aqua). Each satellite scans the entire surface of the Earth every one to two days and passes over China twice a day. The exact timing of passing over China varies with scans and tracks, but is usually between 10 am and 3 pm, and between 9 pm and 2 am in China Standard Time (CST). NASA processes the satellite images and detects fires and other hotspots by thermal anomalies, based on both absolute detection when the fire strength is sufficient to detect and relative detection by comparison with backgrounds to account for variations in surface temperature and brightness. MODIS generally detects fires with a 1 km resolution, although under favorable observing conditions the resolution can be as small as 50 m. NASA takes the routine resolution and reports the detected fires by the center coordinates of the 1 km pixels. For each power plant, we count the total number of fire pixels in a month within a certain distance from the plant as the frequency of fires surrounding the plant in that month. Because the MODIS fire data are available starting from 11 November 2000, we use 2001 to 2019 as our sample period.

Hotspots detected by MODIS include fires in cropland, forests, and grassland, as well as active volcanoes and static land hotspots. To differentiate agricultural fires from other types of hotspots, we use the global land cover type product (MCD12Q1) provided by NASA, which is also derived from MODIS images. This dataset provides yearly land type classifications for each 500 m grid square globally from 2001 to 2019. We match every hotspot in the MODIS fire data to the closest grid square in the MCD12Q1 dataset and identify fires in cropland based on International Geosphere–Biosphere Programme classifications. To avoid the possible influence of fires on land type classifications, we use classifications for the previous year to identify agricultural fires.[15]

[14] The absence of untreated units would result in the under-identification problem of fully-dynamic models in the test for parallel trends (Borusyak et al., 2022). In robustness checks, we also exclude BPPs that are not in operation at the end of the sample period and find that the main results are robust to the exclusion of the untreated plants.

[15] Land type classifications in the previous year are used to identify agricultural fires except for hotspots in 2001. We identify fires in cropland in 2001 based on the land type classifications of the same year because 2001 is the earliest full year available in the MCD12Q1 data.

We obtain the list of registered companies in the biomass power industry from the Registration Information of Industrial and Commercial Enterprises. This dataset contains basic information about companies, including company name, unified social credit code, organization code, registered address, registration date, industry classification, and business scope. Based on company names and business scopes, we retain in our sample companies that generate electricity from direct combustion or gasification of agricultural and forestry waste, and exclude companies that generate electricity only from incineration or gasification of residential garbage and animal waste. In total, we identify 954 registered companies that use agricultural and forestry waste as fuel. To match MODIS fire pixels, we geocode companies by their registered address with *Gaode Maps*, developed by *AutoNavi*, which is a leading online map service provider in China.

To identify the plants that are in operation in the sample, we use the dataset from the National Energy Administration detailing power generation companies with electricity business licenses. The dataset includes company names, unified social credit codes, organization codes, and effective dates and expiration dates of their licenses. We match this dataset with the 954 registered companies by their unified social credit codes and organization codes, and we use the licenses' effective dates as the plants' operational dates. The earliest operation year of BPPs in the sample is 2006, and electricity business licenses have a 20 year duration. This long license duration eliminates the concern that some plants may have renewed their licenses, causing misidentification of their operational dates.

Among the 954 registered companies, 467 began production by the end of 2019. Within the 487 untreated plants, only 43 canceled or revoked their business registration by the end of 2021, implying that most of the untreated plants are likely in the process of preparation rather than withdrawal. The average difference between their registered dates and operational dates is approximately 3 years and 6 months. Fig. D2 in the Online Appendix shows the geographical distribution of the treated and untreated BPPs. Most of the BPPs are located in the east and middle of China, whereas some provinces in the middle and west appear to have a higher proportion of untreated plants. Due to the difference in spatial distributions of treated and untreated plants, comparison of plants across different provinces could be subject to omitted variable biases. This spatial pattern hence highlights the importance of using within-province variations to identify the effects of BPPs.

The control variables for monthly average weather conditions are obtained from the China Meteorological Administration. Data on characteristics of counties where the BPPs are located are taken from county statistical yearbooks.

We consider heterogeneous effects for plants with different liquidity constraints that affect their ability to offer farmers economic incentives for acquiring straw from their land, such as debtor judgments from the supreme court of China[16] and plants receiving the feed-in tariff subsidy disclosed by the Ministry of Finance and the State Grid Corporation. Additionally, we consider the influence of BPPs' demand for straw by comparing the heterogeneous effects of BPPs with different capacity for electricity generation. We use a dataset on national power plants from the China Electricity Council (CEC). This dataset covers national power plants with installed capacity above 6000 kW from 2005 to 2018 and provides plant-level information on generation capacity, electricity generation, and capacity factors.

In order to understand how the entry of BPPs affects farmers' factor inputs in agricultural activities, we use the National Fixed Point Survey conducted by the Ministry of Agriculture on rural households. The survey includes household-level information for more than 20,000 households in 32 provinces from 2000 to 2015, including household income, capital (agricultural machines) and labor inputs for agricultural activities, and land inputs for different agricultural plants. We examine how the entry of BPPs changes rural households' agricultural inputs and income.

We also consider the effects of BPPs on vegetation cover. Following Asher and Novosad (2020), we use two vegetation indices, the normalized difference in vegetation index (NDVI) and the enhanced vegetation index (EVI), to measure local vegetation cover. We use monthly vegetation indices from MODIS Terra with a 1 km resolution (the same size as fire pixels). Further, to examine the effects of BPPs on air pollution, we use annual data on $PM_{2.5}$ concentration provided by the Atmospheric Composition Analysis Group, which uses satellite data on Aerosol Optical Depth and the GEOS-Chem chemical transport model to estimate ground-level $PM_{2.5}$ concentration (van Donkelaar et al., 2019; Hammer et al., 2020).[17]

3.3. Summary statistics

The underlying intuition of our identification strategy is that the untreated plants share similar pre-treatment characteristics with the treated plants, and so can serve as plausible counterfactuals for the treated plants. To confirm this, we examine whether the operating plants are similar to untreated plants within the same province in the pre-treatment period in terms of their location choices.

Table 1 shows the summary statistics for variables in pre-treatment periods. The average monthly fire frequency within 100 km of treated BPPs is 37.317, slightly lower than the frequency around untreated BPPs (41.742). After controlling for province-specific time trends in agricultural fires, the *p*-value of the difference is 0.605, implying no significant difference in pre-treatment agricultural fires between treated and untreated plants.

The availability of biomass resources is a determining factor in site selection for BPPs. Table 1 shows that treated and untreated plants share very similar extents of vegetation cover (NDVI and EVI), implying that their location choices are similar in terms of biomass supply.

In addition, Table 1 shows pre-treatment economic characteristics of counties where the BPPs are located, including population, the value-added of the agriculture sector, industry sector, and service sector, aggregated power of agricultural machines, grain output, and vegetable (including fruits and flowers) output. The counties where treated BPPs are located have larger vegetable output than untreated counties, and the difference is significant at the 5% level. All other variables are very similar between the two groups, and their differences are not statistically significant. The two groups of plants also share similar time-invariant characteristics that may be correlated with agricultural fires, such as the distances to the nearest city center (the seat of city government), to the nearest national highway, and to the nearest airport.

4. Empirical results

4.1. Main results

Fig. 2 presents the coefficients on $Biomass_{iym}$ (circles) in Eq. (1) and their 95% confidence intervals (vertical lines) constructed from Conley spatial standard errors, where the outcome variables are fire frequency in rings binned by a 10 km radius.[18]

[16] The supreme court of China discloses debtor judgments information on the Enforcement Information Disclosure Platform, http://zxgk.court.gov.cn/zhzxgk/.

[17] https://sites.wustl.edu/acag/datasets/surface-pm2-5/

[18] Because the rings farther away from BPPs are larger in area, they tend to have more fire pixels. To compare the coefficients for different rings, we normalize the outcome variables in Fig. 2 based on the average fire frequency in each ring. This normalization preserves the levels of statistical significance and allows us to directly interpret the coefficients as percentage changes in fire frequency.

Table 1
Summary statistics for pre-treatment characteristics.

	Treated			Untreated			*t*-statistics	P-value
	N	Mean	Std	N	Mean	Std		
Fire frequency	72,163	37.317	169.175	111,036	41.742	178.546	0.517	0.605
NDVI	72,163	0.471	0.194	111,036	0.474	0.215	−1.097	0.273
EVI	72,163	0.288	0.141	111,036	0.287	0.152	−0.941	0.347
Population	4266	0.719	0.391	5776	0.650	0.384	0.157	0.875
Agriculture GDP	4935	2.260	1.799	6986	2.569	2.142	0.446	0.656
Industry GDP	4947	5.632	8.126	7012	5.609	9.118	1.335	0.182
Service GDP	4803	3.788	5.288	6755	4.711	9.815	−0.239	0.811
Agricultural machinery	3444	0.539	0.774	4992	0.583	2.837	−1.292	0.197
Grain output	5125	0.667	0.845	7225	0.795	0.979	0.434	0.665
Vegetable output	5125	0.339	0.801	7225	0.255	0.479	2.139	0.033
Distance to city center	467	57.391	48.691	487	67.403	61.282	−0.966	0.334
Distance to highways	467	17.041	27.692	487	16.557	23.238	1.454	0.146
Distance to airports	467	62.777	31.931	487	63.739	35.771	0.930	0.353

Notes: Fire frequency is the monthly frequency of agricultural fires within distance 100 km from BPPs. NDVI and EVI are monthly vegetation indices within distance 100 km from BPPs. Population is the county-level annual population (million). Agriculture GDP is the county-level annual value-added of the agriculture sector (billion CNY). Industry GDP is the county-level annual value-added of the industry sector (billion CNY). Service GDP is the county-level annual value-added of the service sector (billion CNY). Agricultural machinery is the county-level annual power of agricultural machines (gigawatt). Grain output is the county-level annual output of grains (million tons). Vegetable output is the county-level annual output of vegetables and flowers (million tons). Distance to city center is the distance from BPPs to the seats of city governments (km). Distance to highways is the distance from BPPs to the nearest national highways (km). Distance to airports is the distance from BPPs to the nearest airports (km). The *t*-statistics and p-values are constructed after controlling for province-year-month fixed effects for monthly variables, province-year fixed effects for annual variables, and province fixed effects for time-invariant variables.

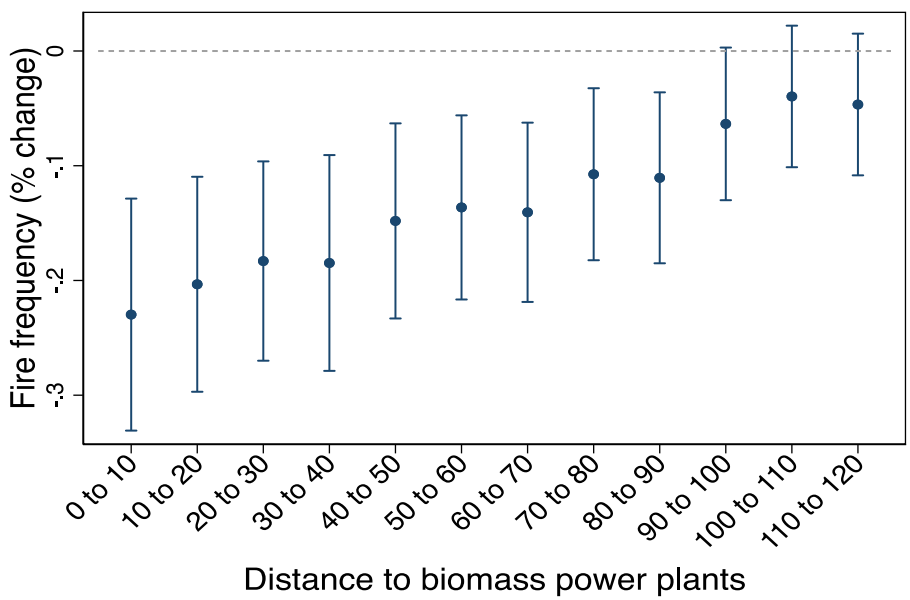


Fig. 2. Effects of biomass power plants on fire frequency by distance. Notes: Circles represent point estimates and lines represent 95% confidence intervals.

All coefficients within the 90 km radius are negative and statistically significant at the 1% level. The effect on agricultural fires in the closest ring is the largest (−23%), whereas fire frequency between 10 km and 40 km of BPPs decreases by more than 18% after BPP entry. The effect on agricultural fires in the rings between 40 km and 70 km is a reduction of about 14%, and the effect in the rings between 70 km and 90 km is a reduction of about 11%. These reductions within the 90 km radius indicate that farmers respond to the entry of BPPs and reduce their use of fires to dispose of straw and clear land.

The coefficients for rings beyond 90 km are not statistically significant at the 5% level, and their magnitudes are much smaller. The decreasing magnitudes by distance from the plants and non-significant effects beyond 90 km highlight the importance of BPPs' transaction costs in reducing agricultural fires. As the distance increases, BPPs bear greater search costs for crop residues and transportation costs to move straw from cropland to their plants. Crop residues are usually large in size but light in weight, creating high transportation costs per ton for crop straw.[19] The high transaction costs reduce the incentive for BPPs to collect straw from faraway cropland. Accordingly, BPP entry does not statistically significantly affect agricultural fires in distant regions, and farmers in those regions continue using fires for land clearing. This finding is consistent with the literature on the importance of transaction costs in the practice of payments for ecosystem services (Jack et al., 2008; Wunder et al., 2020).

Given the non-significant effect on agricultural fires more than 90 km from BPPs, we use the aggregated fire frequency within this distance as an outcome variable and re-estimate Eq. (1). Columns (1)–(3) of Table 2 present the estimation results. Column (1) considers standard two-way fixed effects without any additional controls. The coefficient implies a decrease of 9.27 fires per month after BPP entry relative to the control group. Column (2) uses variations within the

[19] According to Zhao et al. (2017), transportation costs account for more than 20% of operational costs of BPPs.

Table 2
Effects of the entry of biomass power plants on agricultural fire frequency.

	Full sample			High fire seasons			Low fire seasons		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Biomass	−9.270***	−5.075***	−4.836***	−23.372***	−11.471***	−11.192***	0.899	−0.504	−0.428
	(2.621)	(1.370)	(1.326)	(5.774)	(3.200)	(3.106)	(1.107)	(0.484)	(0.470)
Plant	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year-month	Yes	No	No	Yes	No	No	Yes	No	No
Plant-month	No	Yes	Yes	No	Yes	Yes	No	Yes	Yes
Province-year-month	No	Yes	Yes	No	Yes	Yes	No	Yes	Yes
Weather controls	No	No	Yes	No	No	Yes	No	No	Yes
Number of plants	954	954	954	954	954	954	954	954	954
Mean of dep. var	34.428	34.428	34.428	64.286	64.286	64.286	13.101	13.101	13.101
R-squared	0.168	0.709	0.710	0.197	0.700	0.701	0.204	0.705	0.706
Observations	217,512	216,828	216,828	90,630	90,345	90,345	126,882	126,483	126,483

Notes: The dependent variables in Columns (1)–(3), (4)–(6), and (7)–(9) are the monthly frequency of agricultural fires within the 90 km radius of BPPs for all months, for the high fire seasons (March, April, June, October, November), and for the low fire seasons (the rest of months), respectively. Biomass_{iym} is a dummy variable that equals one if plant i is in operation in year y and month m, and zero otherwise. Weather control variables include monthly average temperature, wind speed, relative humidity, and total precipitation. Conley spatial standard errors are in brackets. *, **, and *** indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

same province for identification, by including province-calendar-month fixed effects that absorb any time-varying differences across provinces. It also includes plant-month fixed effects to absorb seasonal variations in nearby agricultural fires for each plant. Consequently, the coefficient in Column (2) is −5.075, which is much smaller than in Column (1), but remains statistically significant at the 1% level. Column (3) further controls for weather conditions. The effect of BPP entry decreases slightly compared with Column (2), but remains highly significant, and suggests that on average BPP entry mitigates agricultural fires within a 90 km radius by 4.863 fires per month, roughly a 14% decrease relative to the average fire frequency of 34.428.

Although the effects are highly statistically significant, the entry of one BPP only mitigates a small percentage of agricultural fires. This result may suggest that transaction costs hinder the economic incentive for farmers to reduce agricultural fires. However, as shown in Fig. 2, even in areas close to BPPs, the entry of BPPs only reduces agricultural fires by about one quarter. Therefore, the quantity of straw collected from farmers who would otherwise burn it is relatively small compared to the substantial demand for straw by BPPs for electricity generation. Consequently, BPPs primarily collect straw from farmers who would have disposed it through alternative means. Most farmers who use fires to clear land would face higher private costs if they were to dispose of straw in an environmentally friendly manner. The entry of BPPs seems to only sufficiently offset such costs for a modest fraction of farmers.

Agricultural fires exhibit strong seasonal variation and closely follow sowing and harvesting activities. To show the seasonal variation in the impact of BPP entry on agricultural fires, we use fire frequency within 90 km as the outcome variable and estimate Eq. (1) for each month. Fig. 3 shows the estimated coefficients (circles) and their 95% confidence intervals (vertical lines). The largest reductions occur in March, April, October, and November. Fig. D3 in the Online Appendix demonstrates that these four months, along with June, have the largest agricultural fire frequencies, aligning with the seasonal patterns of sowing and harvest for major grain crops in China. The coefficients for the other months in Fig. 3 are centered around zero and are of small magnitude, showing no statistical significance.

Therefore, we classify these five months (March, April, June, October, and November) as high agricultural fire seasons, and classify other months as low agricultural fire seasons. We then estimate Eq. (1) for the two groups of seasons separately, with fire frequency within the 90 km radius as the outcome variable. Columns (4)–(9) of Table 2 present the estimation results. For the high fire seasons, Column (6) controls for time-varying province fixed effects and weather conditions, and shows a coefficient of −11.192, significant at the 1% level, suggesting that BPP entry reduces fires by 17% (relative to the 64.286 average fire frequency) in high fire seasons. Coefficients for the low fire seasons in Columns (7)–(9) are much smaller (−0.428 in the last column) and not statistically significant, indicating that BPP entry has no effect in these seasons.

The validity of our event study design relies crucially on the parallel trend assumption. Sun and Abraham (2021) show that in the presence of heterogeneous treatment effects, tests of parallel trends in staggered event study designs with OLS can be contaminated by effects from multiple time periods. To deal with this issue, we examine the parallel trend assumption by using the method proposed by Borusyak et al. (2022) to estimate the following equation:

$$Fire_{iym}^{90} = \sum_{h=-k}^{h=-1} \tau_h \mathbb{I}[K_{iym} = h] + \alpha X_{iym} + \gamma_{im} + \eta_{pym} + \epsilon_{iym} \tag{2}$$

where $Fire_{iym}^{90}$ represents the agricultural fire frequency within a 90 km radius of BPP i in year y and month m, and $\mathbb{I}[K_{iym} = h]$ is the event indicator that equals one for observations h years before the onset of treatment. Other variables are defined as in Eq. (1).

We follow Borusyak et al. (2022) to use only untreated observations to estimate Eq. (2), with observations at least $k + 1$ years before the onset of treatment as the reference group. The choice of k directly affects the estimates on τ_h, and a particularly large k could result in a small reference group and low power of the joint test on τ_h. For robustness, we consider three different values for k, namely 5, 10, and 17. Fig. 4 presents the pre-treatment coefficients, as well as the post-treatment coefficients obtained through an imputation procedure that is not influenced by the choice of k. All pre-treatment coefficients prior to BPP entry are centered around zero and not statistically significant. Notably, the pre-treatment coefficients for $k = 17$ have much wider confidence intervals compared with the coefficients for $k = 5$ and $k = 10$, possibly due to the relatively small reference group when $k = 17$ (i.e., observations 18 years before the onset of treatment). Meanwhile, all post-treatment estimates are negative and most are statistically significant at the 5% level.

One concern for Fig. 4 is that BPP entry may cause a displacement effect on agricultural fires across different months and the effect may contaminate the examination of parallel trends when different months are pooled together, as in Fig. 4. To address this concern, Fig. D4 in the Online Appendix examines the parallel trend assumption for $k = 10$ by each month. For all months, most pre-treatment coefficients are small and not statistically significant. Moreover, most post-treatment coefficients in high fire seasons are negative and statistically significant, but the post-treatment coefficients in other months are not statistically significant, implying no displacement effect.[20]

[20] Some post-treatment coefficients in June are positive but not statistically significant. Moreover, their magnitudes are not large enough to offset the reductions in other months, otherwise the coefficients in Column (3) of Table 2 would not be significantly negative.

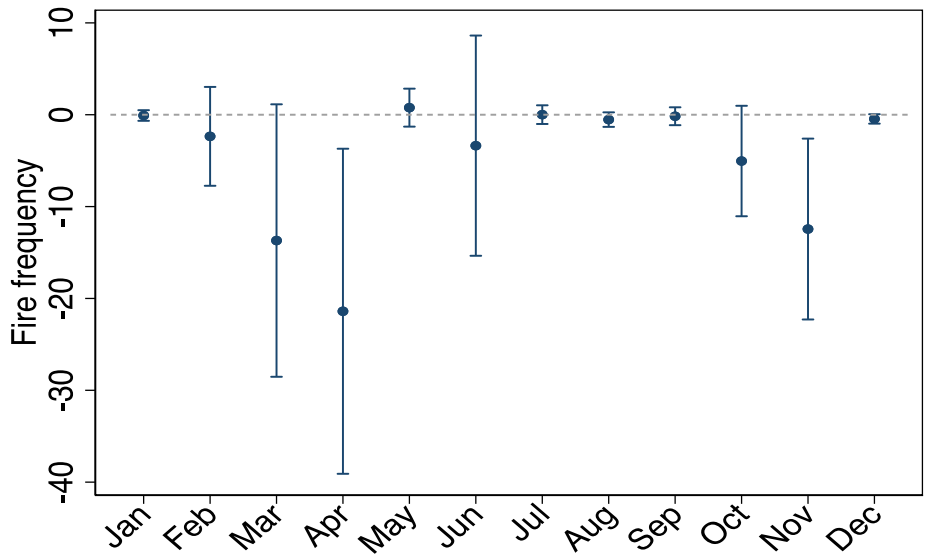
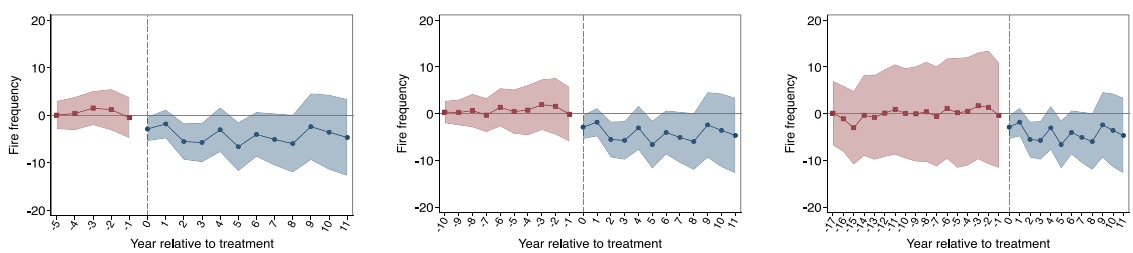


Fig. 3. Effects on fire frequency within a 90 km radius by month. Notes: Circles represent point estimates and lines represent 95% confidence intervals.



(a) $k = 5$ (b) $k = 10$ (c) $k = 17$

Fig. 4. Fire frequency before and after treatment.
Notes: The dependent variables are the monthly frequency of fires within 90 km of BPPs. Square and circles represent point estimates and shaded areas represent 95% confidence intervals.

Table 3
F-tests for pre-treatment trends.

	All months (1)	March (2)	April (3)	June (4)	October (5)	November (6)
	Panel A: $k = 5$					
F-stat	0.466	1.100	0.789	0.345	0.645	0.611
P-value	0.802	0.359	0.558	0.886	0.665	0.692
	Panel B: $k = 10$					
F-stat	0.318	0.999	1.376	0.581	1.207	0.747
P-value	0.977	0.442	0.186	0.830	0.282	0.680
	Panel C: $k = 17$					
F-stat	1.158	0.854	1.518	0.902	1.067	0.740
P-value	0.293	0.630	0.081	0.572	0.382	0.763

Notes: The F-tests proposed by Borusyak et al. (2022) are used to test for pre-trends.

We further follow Borusyak et al. (2022) to formally test for the existence of pre-trends. Specifically, we use an F-test to detect pre-trends for $k = 5, 10, 17$. For all months and for the five months in high fire seasons, Table 3 shows that all 18 F-statistics are quite small and have p-values larger than 0.1, except in April for $k = 17$ (p-value = 0.081) when we have a relatively small reference group. Consequently, the results indicate no significant pre-trends during the pre-treatment years.

4.2. Robustness checks

We perform robustness checks for potential errors in measurement of agricultural fires in this section. In the Online Appendix A, we consider robustness checks for potential measurement errors in plant operation years and locations, as well as for the event-study design.

The MODIS fire product uses thermal anomalies to detect active fires. The algorithm is subject to false alarms caused by reflective surfaces and heated smoke plumes from non-fire sources. However, the fire product reports a confidence estimate, ranging from 0% to 100%, for each detected hotspot, and suggests that users exclude low-confidence hotspots (confidence levels lower than 30%) if they need to reduce potential false alarms. As a robustness test, we follow this suggestion and use hotspots with confidence levels above this threshold for estimations. Omitting low-confidence pixels causes the average number of fires within the 90 km radius to decline from 34.428 to 31.684 per month.

The thermal anomalies detected by MODIS may include hotspots other than vegetation fires, such as active volcanos and other static land sources of heat. In the main results, we use land type classifications to identify hotspots in cropland, which greatly alleviates this concern. To further assure the results are not biased by non-vegetation fires, as a robustness test we exclude all hotspots other than presumed vegetation fires, using the inferred hotspot type for each fire pixel provided by the MODIS fire product as well as the MCD12Q1 land type classifications. This process reduces the number of fires within the 90 km radius to 33.905 per month.

MODIS is equipped on two satellites, Terra and Aqua, which were launched into space in 1999 and 2002. In theory, the two satellites could detect the same fire at different times if the fire occurs at the intersection of the two satellites' orbits, causing overestimation of fire frequency.[21] To avoid possible duplications in counting, we estimate Eq. (1) for hotspots detected by Terra and Aqua separately.[22]

The MODIS fire product routinely detects fires in 1 km pixels. Under favorable observing conditions, the spatial resolution can be much smaller (50 m at the most accurate). To be conservative, we assume that all hotspots are detected at the 1 km resolution, and treat any hotspots within 1 km in the same day as the same fire. We apply the algorithm recursively and estimate Eq. (1) with the re-counted fire frequencies. The average fire frequency per month decreases to 29.032 after applying this algorithm.

Duplications in counting could also occur if a fire persists for more than one day and is detected by satellites on multiple days. As a robustness test, we follow Balboni et al. (2021) to recursively treat hotspots within 1 km in consecutive days as the same fire. This procedure reduces the average fire frequency per month slightly, to 33.024.

Using the six different samples, Fig. 5 presents coefficients and 95% confidence intervals for fire frequency in 10 km binned rings from 0 to 120 km. The results are quite similar to Fig. 2. Coefficients are negative and statistically significant for rings up to 90 km; beyond this distance coefficients are not statistically significant and center around zero. Additionally, conforming with Fig. 2, the effects of BPP entry are generally decreasing in distance from the BPP.

4.3. Mechanisms

In this section, we explore potential mechanisms for the significant effects of BPP entry on reducing agricultural fires. Specifically, we investigate whether these effects are attributable to economic incentives provided by BPPs by examining heterogeneous effects among BPPs with varying abilities to compensate farmers for straw acquisition. We also consider whether the effects are driven by enhanced local enforcement aimed at protecting plant infrastructure, which we test by examining the impact of BPP entry on forest and grass fires. Additionally, we utilize a household-level dataset to investigate whether BPPs influence farmers' decisions on the allocation of their labor and capital inputs.

Economic incentives provided by BPPs. BPPs pay farmers to acquire straw from their land and use the collected straw to generate electricity. If farmers respond to these economic incentives and reduce straw burning, we should expect larger effects from BPPs that are able to provide greater incentives.

To examine this channel, we first consider heterogeneous effects of BPPs between judgment debtors, officially called Persons Subject to Forced Enforcement (PSFEs) and other BPPs. PSFEs are natural persons or legal persons that fail to perform obligations (usually paying off debts) ordered by law courts, and receive restricted access to administrative approval, government support and procurement, loans and financial markets, etc. Being listed as a PSFE indicates that the company lacks adequate liquidity to maintain normal operations. Because straw acquisition costs account for the majority of BPPs' operational costs (Zhao et al., 2017), the BPPs listed as PSFEs are more likely to have difficulties in providing farmers economic incentives to acquire straw. In addition, the lists of PSFEs are disclosed to the public by the Supreme People's Court of China. This public disclosure can directly harm the reputation of PSFE companies and provoke farmers' distrust of the ability or willingness of the BPPs listed as PSFEs to fulfill any pledged economic incentives. Consistent with this conjecture, Jack et al. (2022) show that trust may affect the effectiveness of economic incentives on reducing agricultural fires.

We separately estimate the effects of BPP entry on treated PSFE plants and treated plants never listed as PSFEs, and present the estimation results in Columns (1) and (2) of Table 4, respectively.[23] The coefficient for PSFE plants is 1.780 in high fire seasons and not statistically significant, whereas the coefficient for non-PSFE plants is −14.526, significant at the 1% level. The difference between the two coefficient is 16.306, and the standard error obtained via 100 bootstraps is 3.675, suggesting that the difference is statistically significant at the 1% level. Therefore, in line with our conjecture, PSFE plants experiencing financial liquidity difficulties have a noticeably smaller effect on agricultural fires compared to other plants. We note that the average fire frequencies in high fire seasons for PSFE plants (62.894) and non-PSFE plants (66.076) are similar, indicating that the difference in coefficients is not driven by differences in agricultural fire frequency around PSFE and non-PSFE plants. Farmers around PSFE plants do not significantly reduce their straw burning, not because they use fewer fires to clear land, but it is more likely due to smaller economic incentives provided by PSFE plants or their lack of trust in the promise of payment from PSFE plants.

To promote renewable energy, China began providing a feed-in tariff subsidy to eligible power plants in 2006. The subsidy is not a lump-sum transfer but a form of production subsidy determined by annual electricity generation, so BPPs eligible for the subsidy should be able to provide farmers greater economic incentives to acquire more straw in order to increase their output and subsidy income. The government initially provided a fixed subsidy (0.25 CNY per kWh) to eligible BPPs. In 2010 the subsidy was changed to a fixed on-grid price (0.75 CNY per kWh), which is more than double the average on-grid price for BPPs without the subsidy (0.37 CNY per kWh in 2018). The straw-acquiring cost for BPPs not eligible for the feed-in tariff subsidy must be lower than 0.37 CNY per kWh to break even, whereas BPPs that receive the subsidy can pay up to twice as much to purchase straw while maintaining a positive cash flow.[24] To obtain this subsidy, power plants must apply to their provincial government, and the provincial government then submits applicant lists to the central government (after some preliminary reviews). The central government makes the final decision on whether to approve the application and announces lists of approved plants. We obtain these lists from the websites of the Ministry of Finance and the State Grid Corporation. By the end of 2019, about 60% of treated plants in our sample (271) received the feed-in tariff subsidy.

[21] Terra normally passes over China between 10 am and 1 pm, and between 9 pm and 11 pm (CST), and Aqua normally passes over China between 11 am and 3 pm, and between 1 am and 4 am (CST).

[22] The average numbers of fires within the 90 km radius detected by Terra and Aqua are 19.102 and 18.338, respectively, per month from 2003 to 2019.

[23] More than 80% of PSFE plants in the sample are treated plants. We include all untreated plants (plants not in operation by 2019) in both columns so that we compare each set of treated plants with the same set of untreated plants. We do this consistently in Table 4 because all plants receiving feed-in tariff subsidies by 2019 are treated plants.

[24] Without the subsidy, BPPs' payment for straw is generally less than 200 CNY per ton, but BPPs that receive the subsidy pay more than 300 CNY per ton. Source: https://e.chengdu.cn/html/2013-04/25/content_394366.htm.

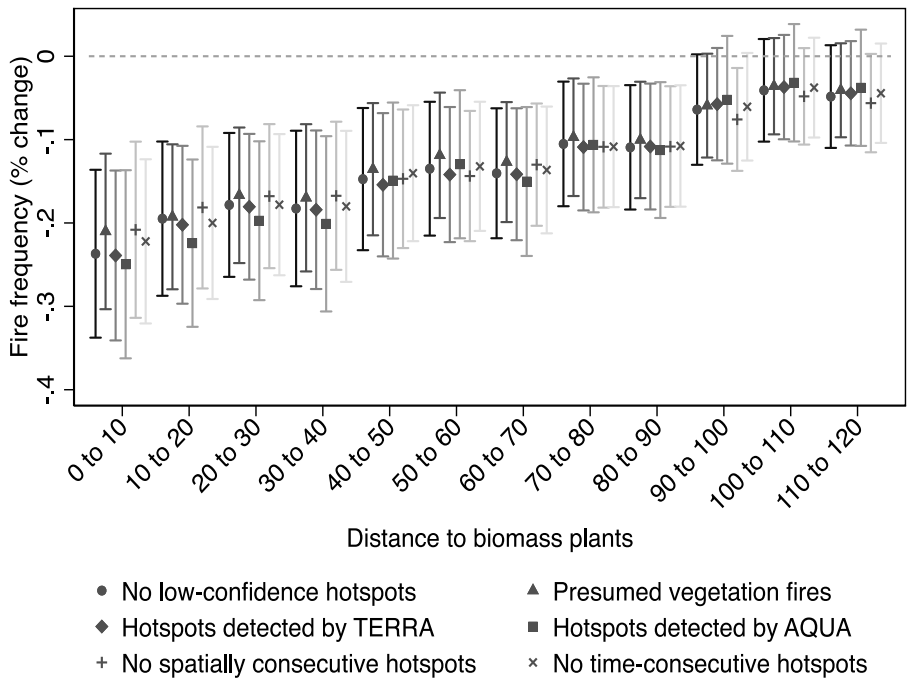


Fig. 5. Robustness checks on measurement errors in fire pixels.
Notes: The dependent variables are the monthly frequency of fires in the ten rings with distance between $r-10$ and r km from BPPs, where $r = 10, \ldots, 120$. Symbols represent point estimates and vertical lines represent 95% confidence intervals.

Table 4
Heterogeneous effects by financial conditions.

	Judgement debtors		Feed-in tariff	
	Yes	No	Yes	No
	(1)	(2)	(3)	(4)
	Panel A: High fire seasons			
Biomass	1.780	−14.526***	−13.948***	−3.181
	(3.780)	(3.604)	(3.516)	(3.596)
Number of plants	592	849	758	683
Mean of dep. var	62.894	66.076	62.122	67.706
Observations	56,145	80,275	71,915	64,505
Difference	16.306*** (3.675)		−10.767* (4.119)	
	Panel B: Low fire seasons			
Biomass	−0.272	−0.458	−0.707	−0.099
	(0.480)	(0.571)	(0.567)	(0.583)
Number of plants	592	849	758	683
Mean of dep. var	12.344	13.066	12.829	12.703
Observations	78,603	112,385	100,681	90,307
Difference	0.186 (0.616)		−0.607 (0.665)	

Notes: The dependent variables in Panel A and Panel B are the monthly frequency of fires within the 90 km radius of BPPs in the high fire seasons and in the low fire seasons, respectively. Columns (1) and (2) represent delinquent plants and non-delinquent plants. Columns (3) and (4) represent plants that do and do not receive feed-in tariff subsidies. All untreated plants are included in the regressions. Biomass_{iym} is a dummy variable that equals one if plant i is in operation in year y and month m, and zero otherwise. Weather control variables include monthly average temperature, wind speed, relative humidity, and total precipitation. Fixed effects are at the plant-month level and province-year-month level. Conley spatial standard errors are in brackets. *, **, and *** indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Columns (3) and (4) of Table 4 show the estimation results for treated plants with and without the feed-in tariff subsidy.[25] The coefficient for subsidized plants is −13.948 in the high fire seasons, significant at the 1% level, whereas the coefficient for plants without the subsidy (−3.181) is only 22.8% of the magnitude and not statistically significant. The difference between the two coefficients is statistically significant at the 10% level. Consistent with our expectation, BPPs that receive the feed-in tariff subsidy have much larger

[25] Table 4 compares plants that have received feed-in-tariff with plants that have never received it. We do not compare the same plants before and after they receive the subsidy. This is because plants that receive the subsidy are typically announced as eligible for feed-in-tariff within two years after they start operating, resulting in a relatively short pre-period to exploit this variation. In addition, once a BPP is deemed eligible for feed-in-tariff by the central government, its electricity production will be compensated for a duration of 15 years, starting from the moment it began grid-connected power generation. If a BPP was included in the provincial government's renewable energy projects development plan when it submitted the project proposal for government approval, it has a fairly high probability of receiving the feed-in-tariff. Consequently, it may provide farmers with significant economic incentives even before the official announcement of subsidy reception.

Table 5
Heterogeneous effects by power generation characteristics.

	Nameplate capacity		Capacity factor		Electricity output	
	High (1)	Low (2)	High (3)	Low (4)	High (5)	Low (6)
	Panel A: High fire seasons					
Biomass	−10.517	−4.246	−41.371**	−1.906	−48.276**	−8.795
	(12.535)	(7.487)	(18.446)	(7.608)	(19.873)	(6.832)
Number of plants	650	627	694	731	673	722
Mean of dep. var	67.824	65.746	67.494	66.705	67.853	66.055
Observations	49,840	50,335	49,895	49,865	49,760	49,710
Difference	−6.272 (15.151)		−39.465* (21.044)		−39.481* (21.294)	
	Panel B: Low fire seasons					
Biomass	0.371	−0.721	−1.150	−0.864	−1.809**	0.474
	(0.857)	(1.063)	(0.790)	(1.131)	(0.871)	(1.206)
Number of plants	650	627	694	731	673	722
Mean of dep. var	12.615	12.753	12.761	12.546	12.693	12.594
Observations	69,776	70,469	69,853	69,811	69,664	69,594
Difference	1.092 (1.240)		−0.286 (1.407)		−2.283* (1.208)	

Notes: The dependent variables in Panel A and Panel B are the monthly frequency of agricultural fires within the 90 km radius of BPPs in the high fire seasons and in the low fire seasons, respectively. Columns (1) and (2) represent biomass power plants with high and low nameplate capacity. Columns (3) and (4) represent biomass power plants with high and low capacity factors. Columns (5) and (6) represent biomass power plants with high and low electricity generation. All untreated plants are included in the regressions. Biomass_{iym} is a dummy variable that equals one if plant i is in operation in year y and month m, and zero otherwise. Weather control variables include monthly average temperature, wind speed, relative humidity, and total precipitation. Fixed effects are at the plant-month level and province-year-month level. Conley spatial standard errors are in brackets. *, **, and *** indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

effects on reducing agricultural fires, implying that farmers are likely to respond more to the potentially greater economic incentives provided by BPPs that receive the production subsidy.

Another factor related to the size of economic incentives is the capacity scale of BPPs, because larger BPPs tend to have greater demand for straw and are likely to offer larger economic incentives to acquire straw. To examine this effect, we use the CEC dataset on national power plants and match our sample with this dataset. From the 467 plants in operation by 2019 per our main sample, we match 296 plants in the CEC dataset. We use the matched treated plants and untreated plants from the original sample for estimations. Columns (1) and (2) of Table 5 examine the heterogeneous effects of BPPs by nameplate generating capacity. The coefficients for BPPs with high and low capacity are −10.517 and −4.246, respectively, in high fire seasons. Nevertheless, neither coefficients are statistically significant due to large standard errors.

BPPs' nameplate capacity may be confounded by unobserved factors related to agricultural fires. To address this concern, Columns (3) and (4) of Table 5 show heterogeneous effects by capacity factors. A capacity factor is a unitless ratio of electricity generation to total capacity, and is likely to be more exogenous to agricultural fire frequency than installed capacity, because capacity factors are determined by local demand for electricity. The effect of BPPs with high capacity factors is quite large (−41.371) in high fire seasons, whereas the effect of BPPs with low capacity factors is only −1.906. The difference between the two coefficients (−39.465) is statistically significant at the 10% level, while the frequency of agricultural fires around BPPs with high and low capacity factors are quite similar (67.494 and 66.705, respectively). The BPPs with higher capacity factors are likely to use more straw to generate electricity and offer larger economic incentives for straw collection. Consistent with this, we observe much larger effects on reducing agricultural fires from the BPPs with higher capacity factors. Columns (5) and (6) provide further support for this channel by showing that BPPs with high annual electricity output have a larger effect (−48.276) in high fire seasons than BPPs with low output (−8.795), and the difference is also significant at the 10% level.

Local enforcement to protect plant infrastructure. Another potential explanation for the significant reduction in agricultural fires is that the entry of BPPs may strengthen local enforcement to protect plant infrastructure against fires.[26] If the observed reduction in agricultural fires is induced by local enforcement on fire prevention, we should expect significant reductions in nearby forest and grass fires, which are usually much more damaging than agricultural fires due to their greater burning intensity and larger burning area.

Fig. 6 shows the effects of the BPP entry on frequency of forest fires and grass fires in the rings binned by the 10 km radius. For both types of fires, all estimates for the rings within 50 km of BPPs are not statistically significant. For rings beyond 50 km from BPPs, two estimates for grass fires are negative and statistically significant but the other five estimates are not significant at the 5% level. Some coefficients for forest fires are even positive and statistically significant, although these have very small magnitudes. The results indicate that there is generally no reduction in the frequency of forest or grass fires around BPPs after they come into operation.

Moreover, if local enforcement is strengthened to protect plant infrastructure against fires, forest and grass fires closer to the plants should decrease more substantially than distant fires because closer fires are more likely to endanger plant infrastructure. Nevertheless, Fig. 6 shows that most coefficients for forest fires remain flat as the distance from BPPs increases. The coefficients for grass fires in the eight rings between 10 km and 90 km also remain relatively flat, with a slightly larger effect for the ring between 50 km and 60 km from BPPs.

Factor reallocation. BPP entry may affect farmers through channels other than economic incentives for acquiring straw. BPPs directly create new jobs for plant operation and straw acquisition. They may also help local industrial development by attracting industrial firms and creating new jobs indirectly. Job creation has an ambiguous impact on agricultural fires. On one hand, farmers may decide to work for BPPs or other industrial firms as part-time workers, and so may reduce their labor input in agricultural activities. The reallocation of labor may induce farmers to use fires as a labor-saving method of clearing their land (Garg et al., 2022). On the other hand, farmers could reduce land input for growing crops or even completely move out of the agriculture sector, which could cause reductions in crop straw and agricultural fires.

To examine these effects, we use household-level data from the National Fixed Point Survey (NFPS) on rural households. We consider how the entry of a BPP within a 90 km radius of households in the NFPS

[26] We thank an anonymous referee for pointing out this potential channel.

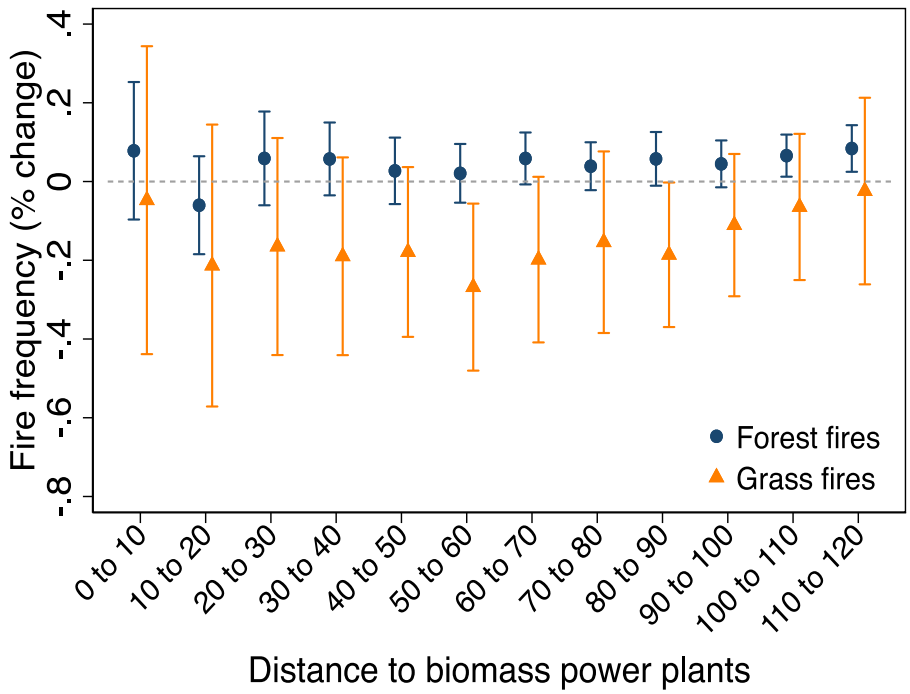


Fig. 6. Effects of the entry of biomass power plants on forest and grass fires.
Notes: The dependent variables are the monthly frequency of forest fires (circles) and grass fires (triangles) in the ten rings with distance between $r-10$ and r km from BPPs, where $r = 10, \ldots, 120$, and the vertical lines represent 95% confidence intervals.

Table 6
Effects of the entry of biomass power plants on rural households.

	Labor input (1)	Planting area (2)	Grain area (3)	Non-grain area (4)	Capital input (5)	Income (6)
	Panel A: OLS estimates					
Biomass	−0.019	145.765	162.111*	−22.289	108.791	3,673.900***
	(1.965)	(127.928)	(94.171)	(64.862)	(449.368)	(868.774)
Number of HH	15,812	16,471	16,468	16,465	14,199	16,463
Mean of dep. var	58	4995	3622	1227	11,283	43,667
Observations	126,232	145,289	145,250	145,195	112,689	145,525
	Panel B: Borusyak et al. (2022) estimates					
Biomass	−0.327	81.832	112.936**	−46.517*	155.484	3,953.136***
	(0.849)	(63.384)	(53.121)	(25.033)	(396.586)	(563.592)
Number of HH	16,283	17,049	17,046	17,043	15,125	17,037
Mean of dep. var	58	5001	3627	1228	11,295	43,617
Observations	126,703	145,867	145,828	145,773	113,615	146,099

Notes: The dependent variables in Columns (1)–(6) are labor input on agricultural activities (days), planting area (square meters), planting area of grains (square meters), planting area of non-grain crops (square meters), agricultural machinery (CNY), and annual household income (CNY). Biomass_{iy} is a dummy variable that equals one if household i is surrounded by an operating biomass power plant within 90 km in year y, and zero otherwise. Fixed effects are at the household level and household type-year level. Conley spatial standard errors (Panel A) and clustered standard errors at the household level (Panel B) are in brackets. *, **, and *** indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

dataset affects their labor, land, and other capital inputs. Table 6 shows the estimation results by the OLS estimators (Panel A) and Borusyak et al. (2022) estimators (Panel B). Column (1) shows the effects of BPPs on the total number of days used for agricultural activities. For both estimators, the effects of BPPs are quite small and not statistically significant, implying that the entry of BPPs does not significantly reduce farmers' labor input in agricultural activities. Column (2) shows the effects of BPPs on total crop planting areas in square meters. The coefficients are positive but not statistically significant, suggesting that the reduction in agricultural fires is not caused by a reduction in land input for growing crops. Instead, Columns 3 and 4 show a statistically significant increase in the area for planting grain crops for both estimators and a smaller decrease in the area for non-grain crops (non-significant for the OLS estimator but significant at the 10% level for the Borusyak et al. estimator). Grain crops are the major source of crop straw used by BPPs to generate electricity, and farmers appear to reallocate their land input towards grain crops that produce more straw, rather than other plants, possibly in response to the economic incentives provided by BPPs to acquire straw that increase the value of grain crops for farmers.

In order to collect straw more efficiently, farmers may also increase capital investment in agricultural machines. It is possible that these machines drive the reduction in agricultural fires. Column (5) shows the effects of BPP entry on machinery used for agricultural activities. The coefficients are quite small (less than 1% relative to the mean) and not statistically significant. Therefore, BPP entry has no significant effect on the capital investment in agricultural machines.

Finally, Column (6) shows the effects of BPPs on household income. If BPPs provide farmers economic incentives to reduce agricultural fires, we should observe that BPP entry is associated with an increase in farmers' income. For both estimators, the effects on household income exceed 3600 CNY and are significant at the 1% level. The average

Table 7
Comparison with the effect of straw-burning bans.

	Poisson			OLS		
	Full (1)	High (2)	Low (3)	Full (4)	High (5)	Low (6)
	Panel A: All fires					
Biomass	−0.154***	−0.189***	−0.063**	−8.505***	−21.006***	0.710
	(0.031)	(0.038)	(0.030)	(2.124)	(4.768)	(0.621)
StrawBans	−0.012	−0.043	0.075**	−4.182*	−8.621	−0.244
	(0.031)	(0.041)	(0.029)	(2.313)	(5.323)	(0.732)
Mean of dep. var	34.428	64.286	13.101	34.428	64.286	13.101
Observations	217,512	90,630	126,882	217,512	90,630	126,882
	Panel B: Daytime fires					
Biomass	−0.162***	−0.202***	−0.066**	−7.657***	−19.130***	0.804
	(0.032)	(0.040)	(0.029)	(1.959)	(4.365)	(0.601)
StrawBans	−0.001	−0.030	0.075***	−4.681**	−10.167**	−0.100
	(0.031)	(0.040)	(0.029)	(1.966)	(4.434)	(0.698)
Mean of dep. var	30.810	56.208	12.668	30.810	56.208	12.668
Observations	217,512	90,630	126,882	217,512	90,630	126,882
	Panel C: Nighttime fires					
Biomass	−0.077	−0.083	−0.034	−0.849**	−1.876*	−0.094
	(0.058)	(0.058)	(0.090)	(0.418)	(0.997)	(0.058)
StrawBans	−0.291***	−0.328***	−0.037	0.498	1.546	−0.144*
	(0.057)	(0.059)	(0.102)	(0.572)	(1.384)	(0.080)
Mean of dep. var	3.618	8.078	0.432	3.618	8.078	0.432
Observations	217,512	90,630	126,882	217,512	90,630	126,882

Notes: The dependent variables in Panels A, B, C are the monthly frequency of all agricultural fires, daytime fires, and night time fires within the 90 km radius of BPPs for all months (Columns (1) and (4)), for the high fire seasons (Columns (2) and (5)), and for the low fire seasons (Columns (3) and (6)), respectively. Biomass_{iym} is a dummy variable that equals one if plant i is in operation in year y and month m, and zero otherwise. Strawbans_{pym} is a dummy variable that equals one if province p completely bans straw burning in year y and month m, and zero otherwise. Weather control variables include monthly average temperature, wind speed, relative humidity, and total precipitation. Fixed effects are at the plant-month level and year-month level. Standard errors are clustered at the plant level. *, **, and *** indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

household income in the sample is about 43,000 CNY, so on average, BPP entry is associated with an 8% increase in household income.

5. Comparison with straw-burning bans

China has officially restricted straw burning for several decades. In 2013, the central government enhanced regulation of straw burning in response to mounting pressure to improve air quality from the national anti-pollution campaign (see Greenstone et al., 2021, for a comprehensive review), and some provincial governments subsequently adopted legislation banning straw burning entirely.[27]

Table D1 in the Online Appendix summarizes the legislation and implementation dates. A total of 15 provinces passed legislation banning straw burning entirely and most started to implement this legislation before 2016. We add a dummy variable $StrawBan_{pym}$ to Eq. (1), denoting whether province p has implemented a complete ban effective in year y and month m. Variation in implementation of straw-burning bans is at the province level, so we cannot use within-province variations to identify the effects of straw-burning bans. We therefore relax the fixed effects in Eq. (1) to use plant-month and year-month fixed effects.

The effectiveness of regulation could be greatly affected by the cost of monitoring agricultural fires, and fires in the nighttime are much easier to spot. The straw-burning bans prohibit burning at all times, including the nighttime. To enforce the bans, government officials carry out some patrols at night, when fires can be detected easily with the naked eye, even from a long distance away. Before 2016, most night fires occurred in June. In our sample, for June in years before 2016, the average night fire frequency within the 90 km radius of BPPs is 45.672, whereas in other months before 2016, the average night fire frequency is only 0.522. However, after 2016, the average frequency of night fires in June declines to 0.916, which is even slightly smaller than the average frequency in other months (1.139).

Table 7 presents the results of Poisson and OLS regressions analyzing the effects of BPP entry and straw-burning bans on agricultural fires occurring at any time, in the daytime, and in the nighttime. Panel A shows the estimation results for fires at any time. The coefficients on $Biomass_{iym}$ estimated by both Poisson and OLS regressions consistently remain negative and highly significant for the full sample period and for high fire seasons. In contrast, the magnitudes of the coefficients on $StrawBan_{pym}$ are much smaller. While the OLS estimator for $StrawBan_{pym}$ in the full period (−4.182) is significantly at the 10% level, its magnitude is considerably smaller than the effect of BPP entry (−8.505). Moreover, the Poisson estimators for $StrawBan_{pym}$ are not significant for the full period and high fire seasons (−0.012 and −0.043), and even significantly positive during low fire seasons (0.075), suggesting the possible occurrence of inter-temporal displacement. These results indicate that complete bans on straw burning have a substantially smaller effect on agricultural fires compared to the effect of BPP entry.

Panels B and C of Table 7 present the effects of BPP entry and straw-burning bans on agricultural fires in the daytime and nighttime. For daytime fires in the full sample period and high fire seasons, both the Poisson and OLS estimators show negative and statistically significant coefficients on BPP entry at the 1% level. For nighttime fires, the effects remain negative but not statistically significant for the Poisson estimators. While the OLS coefficient on straw-burning bans is significantly negative for daytime fires in high fire seasons (−10.167), its magnitude is still much smaller than the effect of BPP entry (−19.130), and the Poisson coefficient remains not statistically significant in high fire seasons (−0.030) and significantly positive in low fire seasons (0.075).

On the other hand, the Poisson coefficients on straw-burning bans for nighttime fires are −0.291 and −0.328 for the full sample period

[27] Previous regulation generally targeted certain areas and periods, whereas the complete ban prohibits any straw burning in all areas of the province throughout the year.

and high fire seasons, both significant at the 1% level.[28] The coefficients suggest a roughly 30% decline in night fires due to the straw-burning bans. Therefore, although the bans appear to have little effect on mitigating agricultural fires during the daytime, they effectively reduce fires in the nighttime, when fires are easier to detect. Nevertheless, agricultural fires in the nighttime are much less frequent than fires in the daytime; the average fire frequency in the nighttime is only 3.618 for the full sample period, whereas the average frequency in the daytime is 30.810. Therefore, despite the significant effect on reducing nighttime fires, the straw-burning bans have a negligible effect on reducing total number of agricultural fires.

In the Online Appendix B, we further explore the inter-temporal displacement effects of the straw-burning bans on agricultural fires by estimating the effects by each month. Our findings indicate that the bans effectively reduce fires in June, September, and November, but fires rebound in other months. We show that this pattern appears to align with variations in the enforcement stringency of straw-burning bans.

6. Benefits and costs

6.1. Implications for air pollution and health

Reductions in fires due to BPP entry imply improvements in air quality, but the combustion of straw in BPPs' generators will also produce air pollutants, such as SO_2, NO_X, and $PM_{2.5}$. However, BPPs in China face strict regulation of emissions of air pollutants. They are required to install filters, such as dust collectors and desulfurizing and denitrifying facilities, to achieve emissions standards. Therefore, air pollutant emissions from BPPs should be considerably lower than emissions from open burning of the same quantity of straw in farmland. According to the emissions inventory for biomass burning compiled by the Ministry of Environmental Protection of China, open burning generates 6.79 g of $PM_{2.5}$ and 6.93 g of PM_{10} per kilogram of straw, whereas biomass steam boilers with standard wet dust collectors generate 0.48 g of $PM_{2.5}$ and 0.49 g of PM_{10}. If they are equipped with highly efficient bag filters, $PM_{2.5}$ and PM_{10} emissions will be further reduced to 0.05 and 0.06 g, about 0.8% of the emissions from open burning. Even biomass steam boilers without any filters generate 0.95 g of $PM_{2.5}$ and 1.12 g of PM_{10} per kilogram of straw, only 14% and 16% of the emissions from open burning, respectively. These estimates imply that for each unit reduction in agricultural fires due to BPP entry, particulate matter emitted from the fires is reduced by at least 84% even if no filters are installed by the BPPs.

We also provide empirical evidence of the improvements in air quality by using satellite-based annual data on $PM_{2.5}$ concentration from 2001 to 2018 to examine the effect of BPP entry on local $PM_{2.5}$ concentration. Panel A of Table 8 shows the OLS estimation results. We consider three different radii—30 km, 50 km, and 90 km—and calculate average concentrations within the radii. All coefficients on BPP entry are negative and statistically significant at the 5% level. BPP entry is associated with a 0.917 $\mu g/m^3$ reduction in average $PM_{2.5}$ concentration within the 30 km radius (Column (1)), about 2% of the average annual concentration. As a robustness test, we also use the estimation method proposed by Borusyak et al. (2022) and show the estimation results in Panel B of Table 8. Although the magnitudes of coefficients are slightly smaller, they are all statistically significant at the 1% level.

Fig. D5 in the Online Appendix further shows the effects of BPPs on average concentration in the rings binned by a 10 km radius. Interestingly, although we observe no statistically significant reduction in agricultural fires beyond 90 km from BPPs (Fig. 2), Fig. D5 shows that the reduction in $PM_{2.5}$ concentration is statistically significant at the 5% level for rings up to 120 km to BPPs. This effect is most likely attributable to spatial spillover of air pollutants. The magnitudes of the coefficients in Fig. D5 are also much more similar than those in Fig. 2, suggesting that the pollutant concentrations in these rings are highly correlated.

The significant reduction in fire frequency and improvement in air quality imply large health benefits to rural residents. Column (3) of Table 2 suggests that the entry of one BPP reduces monthly agricultural fires by 4.863. Based on the estimated health damage caused by agricultural fires calculated by He et al. (2020),[29] and the average mortality at the county level,[30] the annual reduction in mortality due to the 467 plants in our sample amounts to 12,803 lives ($4.836/10 \times 0.0156 \times 3634 \times 467$). If we take the value of a statistical life of 2.92 million CNY, as used by Fan et al. (2020) and He et al. (2020), the annual health benefits from the reduction in mortality are about 37 billion CNY.

Fig. 7 illustrates that the health benefits resulting from improved air quality due to reduced agricultural burning are substantial, even in comparison to the social benefits derived from reducing carbon emissions. The biomass power sector generated 48.1 billion kWh of electricity in 2019. Because the Chinese government commits to purchasing all electricity generated by renewable energy sources including biomass, and more than 90% of thermal electricity in China is generated from coal during the sample period, we assume that all electricity generated by BPPs substitutes for electricity from coal-fired plants. Then the reduction in carbon emissions amounts to 28.379 million tons, which is a substantial decrease.[31] In fact, this reduction is more than twice the carbon emission reduction attributable to high-speed rail (Lin et al., 2021). Assuming a social cost of carbon in China of 26 USD per ton (Ricke et al., 2018), the reduction in carbon emissions can be valued at about 5 billion CNY. However, this value is only 14% of the monetized health benefits derived from reductions in agricultural fires. Therefore, in addition to the substantial benefits from reducing carbon emissions, BPPs in China deliver much greater health benefits by mitigating agricultural fires.[32]

Furthermore, Fig. 7 shows that the health benefits to the public exceed BPPs' private revenues from selling the generated electricity. The National Administration of Energy disclosed that the average on-grid electricity price is 0.67799 CNY/kWh for BPPs in China in 2018.[33] Therefore, the total revenues from selling the 48.1 billion kWh of electricity amount to 33 billion CNY, which is about 10% smaller than the health benefits.

[28] Interestingly, the OLS coefficients are positive but not statistically significant. Given that about 83% of the observations in the full sample period have no nighttime agricultural fires (and fire frequency is a count variable), it is more appropriate to model the heavy-tailed distribution using an exponential model rather than a linear model. In addition, the fixed-effects Poisson model demonstrates high robustness (Wooldridge, 1999), even in the difference-in-differences setting (Wooldridge, 2022). Therefore, we consider the results of the Poisson regression to be more reliable than those of OLS for nighttime fires.

[29] He et al. (2020) find that an increase of 10 fires in the number of fires leads to a 1.56% increase in monthly deaths from all causes.

[30] The average annual mortality per county within the 90 km radius of a BPP was 3634 in 2019, according to the China City Statistical Yearbook and county-level China Statistical Yearbook.

[31] The median of lifecycle carbon emission intensities of coal-fired and biomass-fired electricity are 820 and 230 g/kWh, respectively, according to Intergovernmental Panel on Climate Change (2015). The reduction in carbon emissions is $48.1 \times (820-230)/1000 = 28.379$ million tons.

[32] The social cost of carbon is a contentious topic, with estimates ranging from dozens to hundreds of USD per ton. However, the large relative difference between the health benefits and benefits from reduction of carbon emissions implies that the two benefits would be equivalent only under a high estimate for the social cost of carbon (about 186 USD per ton).

[33] Source: http://www.nea.gov.cn/138530255_15729388881531n.pdf.

Table 8
Effects of the entry of biomass power plants on $PM_{2.5}$.

	$r \leq 30$ (1)	$r \leq 50$ (2)	$r \leq 90$ (3)
	Panel A: OLS estimates		
Biomass	−0.917**	−0.891**	−0.837**
	(0.397)	(0.393)	(0.381)
Number of plants	954	954	954
Mean of dep. var	46.714	45.974	45.051
Observations	17,172	17,172	17,172
	Panel B: Borusyak et al. (2022) estimates		
Biomass	−0.853***	−0.832***	−0.778***
	(0.314)	(0.309)	(0.296)
Number of plants	954	954	954
Mean of dep. var	46.714	45.974	45.051
Observations	17,172	17,172	17,172

Notes: The dependent variables in Columns (1)–(3) are the annual average $PM_{2.5}$ concentrations within 30, 50, and 90 km to BPPs, respectively. $Biomass_{iy}$ is a dummy variable that equals one if plant i is in operation in year y, and zero otherwise. Weather control variables include annual average temperature, wind speed, relative humidity, and total precipitation. Fixed effects are at the plant level and year level. Conley spatial standard errors (Panel A) and clustered standard errors at the plant level (Panel B) are in brackets. *, **, and *** indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

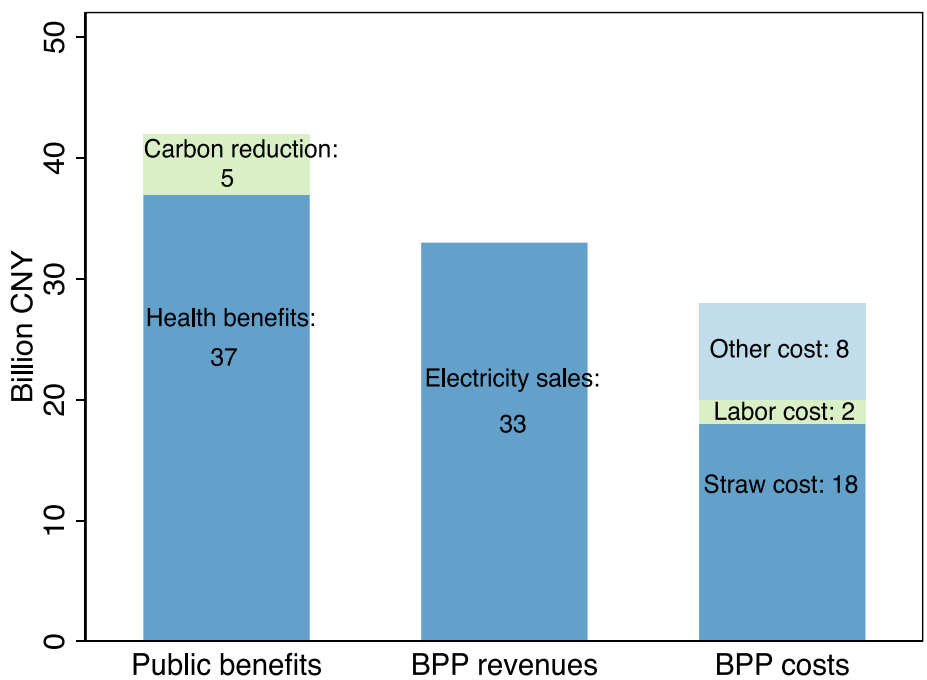


Fig. 7. Annual benefits and costs of biomass power plants.

6.2. Costs of biomass power plants

Fig. 7 also shows operational costs of BPPs, based on estimates from Zhao et al. (2017).[34] More than 60% of the costs are for acquiring straw (18 billion CNY), representing payments to farmers. BPPs' labor costs and other costs (including costs of transporting straw) are 2 billion CNY and 8 billion CNY, respectively. The total operational costs of BPPs amount to 28 billion CNY, leaving 5 billion annual profits from which to recover their investment costs.

A great concern about using biomass to generate electricity is that it could reduce forests and threaten biodiversity.[35] However, this criticism addresses use of forests for bioenergy, whereas BPPs in China mainly use agricultural waste as fuel. Furthermore, most forests in China are state owned or collectively owned, and the government has implemented strict regulation of woodland for several decades. Large-scale logging without government permission is strictly prohibited.[36]

Due to the nature of forest ownership and strict regulation, we expect the effects of BPP entry on loss of vegetation cover to be limited. To test this conjecture, we examine the effects of BPP entry on two vegetation indices (NDVI and EVI). Similar to Table 8, we consider three different radii (30, 50, and 90 km) and calculate the logarithms of average indices within the radii. Table 9 shows the estimation results.

[34] Zhao et al. (2017) find that the straw cost for biomass power plants is 0.382 CNY/kWh, labor cost is 0.042 CNY/kWh, and other costs are 0.175 CNY/kWh.

[35] Over 500 scientists wrote to the Presidents of the EU Commission, the US, and South Korea, and the Prime Minister of Japan in February 2021, urging the governments to stop subsidizing wood burning for electricity. Source: https://www.wwf.eu/?2128466%2F500-scientists-tell-EU-to-end-tree-burning-for-energy.

[36] According to the forest law of China, before cutting down more than two trees, even if the trees are on a farmer's own land, the farmer must submit a formal request to local government and obtain a forest logging permit.

Table 9
Effects of the entry of biomass power plants on vegetation indices.

	NDVI			EVI		
	$r \leq 30$	$r \leq 50$	$r \leq 90$	$r \leq 30$	$r \leq 50$	$r \leq 90$
	(1)	(2)	(3)	(4)	(5)	(6)
	Panel A: OLS estimates					
Biomass	0.001	0.002	0.004**	−0.003	−0.002	0.000
	(0.003)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
Number of plants	954	954	954	954	954	954
Mean of dep. var	−0.927	−0.909	−0.887	−1.432	−1.420	−1.406
Observations	216,695	216,749	216,808	216,666	216,718	216,792
	Panel B: Borusyak et al. (2022) estimates					
Biomass	0.000	0.001	0.003	−0.004	−0.003	−0.001
	(0.003)	(0.003)	(0.002)	(0.003)	(0.003)	(0.002)
Number of plants	954	954	954	954	954	954
Mean of dep. var	−0.927	−0.909	−0.887	−1.432	−1.420	−1.406
Observations	217,361	217,415	217,474	217,332	217,384	217,458

Notes: The dependent variables in Panel A and Panel B are monthly average vegetation indices (NDVI for Columns (1)–(3) and EVI for Columns (4)–(6)) for the high fire seasons and for the low fire seasons, respectively. Columns (1) and (4) represent average indices within 30 km of BPPs. Columns (2) and (5) represent average indices within 50 km of BPPs. Columns (3) and (6) represent average indices within 90 km of BPPs. Biomass_{iym} is a dummy variable that equals one if plant i is in operation in year y and month m, and zero otherwise. Weather control variables include monthly average temperature, wind speed, relative humidity, and total precipitation. Fixed effects are at the plant-month level and province-year-month level. Conley spatial standard errors (Panel A) and clustered standard errors at the plant level (Panel B) are in brackets. *, **, and *** indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

For both OLS estimators (Panel A) and Borusyak et al. (2022) estimators (Panel B), we find that the coefficients on BPP entry for NDVI and EVI are quite close to zero and none are statistically significantly negative. One coefficient is even statistically significantly positive, although small in magnitude (0.004). These results indicate that BPPs in China exert little influence on reducing local vegetation cover.

7. Conclusion

This paper provides empirical evidence of the effect of BPP entry on reducing agricultural fires, a prevalent method used by farmers in many countries to clear their land. Exploiting the entry of BPPs as a quasi-natural experiment, we show that BPPs have, on average, a statistically significant effect on reducing straw burning. Within a 90 km radius of a new BPP, agricultural fires reduce by 14%. Moreover, we find that farmers closer to BPPs exhibit a larger response (up to −23%), with the effect diminishing gradually as the distance from the BPP increases. This implies a critical role of BPPs' transaction costs in internalizing environmental externalities. The effects of BPP entry are also much stronger in months before sowing and after harvesting (−17%) than other months (−3%).

We show that the reduction in agricultural fires is likely induced by economic incentives provided by BPPs to farmers for acquiring straw from their land, because BPPs that are able to provide larger economic incentives to farmers exhibit greater effects on reducing nearby agricultural fires. We also show that farmers appear to respond to the economic incentives by allocating more land to growing grain crops, which produce more straw for BPPs to acquire.

We further compare the effects of BPPs with the effects of straw-burning bans. Agricultural fires are generally much smaller than forest fires and harder to monitor. We find that straw-burning bans have limited effects on reducing agricultural fires. However, the bans appear to significantly reduce straw burning at night, when monitoring of agricultural fires is much easier than during the day.

Our back-of-the-envelope calculations indicate that public health benefits from improved air quality due to the reduction in agricultural fires are considerably larger than the monetized social benefits derived from carbon emission reductions achieved through the substitution of coal-fired electricity with electricity generated by BPPs. Moreover, these public health benefits are slightly larger than BPPs' private revenues from electricity sales. These substantial health benefits imply that BPPs, often recognized as (controversial) contributors to greenhouse gas emission reduction and climate change mitigation, play an even more significant role in reducing local air pollutants and improving human health. It is likely that similar health benefits exist in other approaches adopted by countries to reduce their greenhouse gas emissions.

At present, the opportunity cost of labor is rising rapidly in many developing countries. The rising labor cost could directly increase BPPs' straw-acquiring costs and aggravate their financial difficulties, especially under a regulated electricity price. Whether BPPs should, in future, be compensated more for the public health benefits they provide, or whether we should seek a more cost-effective way to control agricultural fires, merits further investigation.

CRediT authorship contribution statement

Jing Cao: Methodology, Resources, Writing. **Rong Ma:** Data curation, Methodology, Formal analysis, Writing.

Data availability

The data is available online as supplementary material at the following link: https://doi.org/10.1016/j.jdeveco.2023.103173.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.jdeveco.2023.103173.

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